

# Raj Kolala

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## Education

### San Jose State University

May 2027

Bachelor of Engineering, Computer Networking System Management, Minor in Business Administration

San Jose, CA

Relevant Coursework: Network Administration, Intro to Engineering, Intro to Python, Physics II, Analog and Digital Circuits, Intro to IoT, Intro to Networks, Calculus, Linear Algebra, Public Speaking, Business Statistics

## Technical Skills

Languages: AWS, Azure, GCP, Python, PowerShell, Bash, Docker, Kubernetes, Postman, Linux, VMWare, PuTTY, Wireshark, Cisco Equipment, TCP/IP Protocols, DNS, DHCP, HTTP/HTTPS, SSH, FTP, VPN, VLAN, ISO 27001, React Native, Node.js, WebSocket

## Experience

### Cisco

July 2023 – Nov 2023

Network Engineering Consultant

Santa Clara, CA

- Collaborated with **150+ IT professionals and educators** to deliver workshops on **data literacy, cybersecurity, and generative AI integration**, configuring practical solutions for real-world business use cases.
- Designed and facilitated breakout sessions improving understanding of **AI-driven productivity tools** and ethical implications — feedback surveys showed a **92% satisfaction rate** across **20+ organizations**.
- Supported consulting sessions diagnosing **data anomalies and security vulnerabilities** in large-scale enterprise systems, applying **network defense principles** to assess risk exposure across distributed web infrastructures.

## Projects

### Clarity – See Yourself | React Native, Expo, WebSocket, Node.js, EAS Build

Jan 2025 – Present

- Architected a **real-time peer-to-peer video mirroring system** for iOS using **WebSocket protocol** over a **Node.js relay server**, enabling sub-second latency camera feed synchronization between two devices via a **6-character room code session handshake**.
- Developed and shipped a **full-stack iOS application** to the App Store integrating a **client-server WebSocket architecture**, live **camera stream processing** with 14 real-time editing parameters, and local on-device storage with **zero data transmission to external servers**.
- Designed and deployed a **companion support website** with a live **privacy policy endpoint** at [clarityseeyourself.com](https://clarityseeyourself.com).

### End-to-End Network Prototype | Cisco Packet Tracer, TCP/IP, Routing, DNS

Oct 2025 – Dec 2025

- Architected and simulated an internet-scale, **multi-tier ISP topology** connecting **home, SMB, and cellular access networks** to a centralized data center through hierarchical upstream transit.
- Engineered **Tier-3, Tier-2, and Tier-1 ISP interconnections**, implementing end-to-end IP addressing, routing, and DNS to validate packet traversal and scalability across **15+ routed networks**.

### Integrated IoT Environment | Python, Ubuntu, Docker, Zigbee

Nov 2024 – Apr 2025

- Engineered a **smart home ecosystem** integrating **15+ IoT devices** (Zigbee & Z-Wave) via **Home Assistant and Docker-based microservices**, achieving **95% uptime** with automated control of lighting, temperature, and security.
- Configured and deployed a **custom Home Assistant instance on Raspberry Pi 5**, optimizing resource utilization and creating modular automation scripts in **Python** to streamline device discovery and voice control.

## Certifications

### Google: Cybersecurity Professional Certificate | SQL, Linux, SIEM Tools

November 2025

- Completed a 9-course program emphasizing **enterprise-level** threat detection, incident response, and network defense; applied **NIST, MITRE ATT&CK, and OWASP** frameworks for vulnerability assessments, log analysis, and full-cycle incident remediation.

### AWS Academy: Cloud Architecting | EC2, S3, RDS, DynamoDB, Lambda, VPC

November 2025

- Completed cloud architecture coursework covering **VPC design**, compute and storage services (EC2, Lambda, S3), IAM governance, load balancing, and best practices for **reliability, scalability, and cost optimization**.

### CCNA: Enterprise Networking, Security, and Automation | Packet Tracer, Ubiquiti

March 2025

- Designed and simulated **enterprise-grade network architectures** using advanced **IPv4/IPv6 routing and Layer 3 switching protocols**, achieving **100% connectivity and redundancy** in complex topologies.

### CCNA: Switching, Routing, and Wireless Essentials | Cisco Networking Academy

June 2024

- Configured and optimized **Layer 2/Layer 3 switches, routers, and wireless LANs** for small-to-medium business infrastructures with **zero packet loss** under test loads.